NOTICE 1
23 October 2012

TEST OPERATIONS PROCEDURE

This Test Operations Procedure (TOP) has been archived per ATEC memorandum dated 18 October 2012.

Requests shall be referred to Headquarters, U.S. Army Test and Evaluation Command, 2202 Aberdeen Boulevard, Aberdeen Proving Ground, MD 21005-5055, usarmy.APG.atec.mbx.atec-standards@mail.mil

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REFERENCE FILE

25 September 1969

U. S. ARMY TEST AND EVALUATION COMMAND
COMMODITY ENGINEERING TEST PROCEDURE

NUCLEAR YIELD MEASURING DEVICES

1. OBJECTIVE

The objective of this materiel test procedure is to describe the engineering tests required to determine the technical performance and safety characteristics of nuclear yield measuring equipment and their associated maintenance package relative to criteria contained in applicable Qualitative Materiel Requirements (QMR), Small Development Requirements (SDR), Technical Characteristics (TC), and other appropriate technical specifications, and determining their suitability for military use.

2. BACKGROUND

Nuclear warfare operations prompted the development of nuclear yield equipment. A Department of the Army approved QMR was issued in February 1964 for this type of equipment.

The principle application of a nuclear yield measurement device is to predict the amount of radioactive fallout from nuclear detonations. Basically, such a device is designed to provide valid and timely nuclear burst yield information, and record the yield of a nuclear event.

The present ban on atmospheric nuclear weapons testing imposes special consideration in the test methods used in the conduct of the engineering test of a nuclear yield measuring device.

3. REQUIRED EQUIPMENT

a. Photographic equipment
b. Oscilloscope
c. Signal generator
d. Voltmeter
e. Ammeter
f. Environmental test facility
g. Linear accelerator (LINAC)
h. Safety equipment, including appropriate dosimeters
i. Calorimeters
j. Radioactive sources
k. EMP (Electromagnetic Pulse) generator
l. Electromagnetic illuminator

4. REFERENCES

A. MIL-STD-810-B, Environmental Test Methods
B. MIL-STD-461, Electromagnetic Interference Characteristics Requirements for Equipment

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C. MIL-STD-462, Electromagnetic Interference Characteristics, Measurement of
D. MIL-STD-463, Definitions and Systems of Units, Electromagnetic Interference Technology
E. AR 705-15 (C-1), Operation of Materiel Under Extreme Conditions of Environment
F. AD No. (to be assigned), STP Susceptibility Evaluation, Report of Task performed for U. S. Army Mobility Equipment Research and Development Center, Ft. Belvoir, Virginia, under Contract No. DAAK02-67-C-0168. January 1969 (Secret Restricted Data)
G. DASA, EMP (Electromagnetic Pulse) Handbook, Prepared for Defense Atomic Support Agency under Contract No. DASA 01-68-C-0007 by General Electric Company, September 1968 (Unclassified)
H. DASA, EMP (Electromagnetic Pulse) Handbook, Prepared for Defense Atomic Support Agency under Contract No. DASA 01-68-C-0007 by General Electric Company, September 1968, (Classified Supplement)
I. DASA 1420, TREE (Transient-Radiation Effects on Electronics Handbook, Battelle Memorial Institute, Columbus, Ohio, August 1967
J. DASA 1420, TREE (Transient-Radiation Effects on Electronics) Handbook, Classified Supplement DASA 1420-1, July 1966 (Secret Restricted Data)
K. Qualitative Materiel Requirement (QMR), Nuclear Yield Measurement Equipment, 24 February 1964. (Confidential)
L. Chalmers, J. Alan, Atmospheric Electricity, Second Edition, Pergamon Press, 1967
M. The Effects of Nuclear Weapons, U. S. Atomic Energy Commission 1964
N. Terman, F. E., and Pettit, J. M., Electronic Measurements Second Edition, McGraw-Hill Book Co., Inc., 1952
O. AMCR 385-25, Radiation Protection
P. TB MED 270, Control of Hazards to Health from Microwave Radiation
Q. MTP 3-1-002, Confidence Intervals and Sample Size
R. MTP 5-2-584, Microbial Resistance Tests
S. MTP 6-2-507, Safety
T. MTP 7-2-509, Air Drop Capability of Materiel

5. SCOPE

5.1 SUMMARY

5.1.1 Technical Characteristics

The specific commodity sub-tests to be performed are listed below:

a. Response requirements - The objective of this sub-test is to: (1) establish functional characteristics control data for the test unit, and (2) assess the unit performance degradation, both during and after completion of the environmental tests.

b. EMC-RFI - The objective of this sub-test is to determine the

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ability of the test item to be operationally compatible with the RF environment, both radiated and received, that is likely to be encountered if the item is fielded in accordance with the QMR.

c. SFERICS response - The objective of this sub-test is to determine capability and reliability of the test item to discriminate and reject natural atmospheric electricity and phenomena associated with thunderstorm activity.

d. Nuclear weapons effects - The objective of this sub-test is to determine the following specific effects on the test item in a time sequence: (1) Gamma radiation; (2) Neutron radiation; (3) Fast burst reactor effects; (4) Thermal radiation; and (5) Air blast.

e. Microbiological test - The objective of this sub-test is to determine the resistance of the test item to microbiological attack.

f. Battery Load and Shelf Life Test - The objectives of this sub-test are to determine: (1) the in-service performance characteristics while under load and while under various temperature environments; and (2) the shelf life of fully activated batteries.

g. Equipment reliability - The objectives of this sub-test are to determine: (1) length of time of continuous operation at the prescribed reliability; (2) battery life at prescribed reliability; (3) false alarm rate; and (4) life time probability distribution of entire equipment.

h. Air drop - The objective of this subtest is to determine the extent system or subsystem performance is degraded as a result of field delivery by air drop methods.

5.1.2 Common Engineering Tests

The following Common Engineering Tests, applicable to these commodities, are not included in this MTP:

a. 6-2-500, Physical Characteristics
b. 6-2-502, Human Factors Engineering
c. 6-2-503, Reliability
d. 6-2-504, Design for Maintainability
e. 6-2-530, Altitude and Temperature-Altitude Tests
f. 6-2-531, Temperature Tests
g. 6-2-532, Sunshine Tests
h. 6-2-533, Rain Tests
i. 6-2-534, Humidity Tests
j. 6-2-536, Salt Fog Tests
k. 6-2-537, Dust Tests
l. 6-2-539, Immersion Tests
m. 6-2-540, Vibration Tests
n. 6-2-541, Shock Tests

5.2 LIMITATIONS

The variety of nuclear yield measuring devices to which this MTP is applicable precludes detailed coverage of any particular item. The testing methods outlined are intentionally general to provide test coverage for various

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instruments and may be adapted, as necessary to accomodate specific equipment.

6. PROCEDURES

6.1 PREPARATION FOR TEST

a. Select test equipment ideally having an accuracy of at least ten orders of magnitude greater than that afforded by the item under test, that is in keeping with the state of the art, and with calibrations traceable to the National Bureau of Standards.

b. Record the following information:

1) Nomenclature, serial number(s) manufacturer's name, and function of the item(s) under test.
2) Nomenclature, serial number, accuracy tolerances, calibration requirements, and last date calibrated of the test equipment selected for the tests.

c. Ensure that all test personnel are familiar with the required technical and operational characteristics of the item under test, such as stipulated in Qualitative Materiel Requirements (QMR), Small Development Requirements (SDR), and Technical Characteristics (TC).

d. Review all instructional material issued with the test item by the manufacturer, contractor, or government, as well as reports of previous similar tests conducted on the same types of equipment. These documents shall be kept readily available for reference.

e. Prepare record forms for systematic entry of data, chronology of test, and analysis in final evaluation of the test item.

f. Prepare adequate safety precautions to provide safety for personnel and equipment, and ensure that all safety SOP's are observed throughout the test and that the item has successfully completed the examination prescribed in MTP 6-2-507, Safety.

NOTE: 1. All radiological safety requirements in AMCR 385-25 shall be complied with.
2. Test personnel working with radioactivity for the first time shall be informed of the radiation hazards associated with their work and their right to review radiation exposure records indicating their accumulated dose.
3. High voltage in most of these instruments require cautious handling during maintenance to prevent shock; many instruments, though switched to "OFF" position, still retain sufficient plate capacitance charge to create a shock hazard.
4. Following receipt, each radioactive source shall be leak tested in accordance with AEC and DA directives.

g. Thoroughly inspect the test item for obvious physical and electrical defects such as cracked or broken parts, loose connections, bare or broken wires, loose assemblies, bent fragile parts, and corroded plugs and jacks.

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All defects shall be noted and corrected before proceeding with the test.

h. Prior to beginning any sub-test, verify that the equipment is aligned, if necessary, as specified in the pertinent operating instructions to ensure, insofar as possible, it represents an average equipment in normal operating condition.

i. Prepare a test item sample plan sufficient to ensure that enough samples of all measurements are taken to provide statistical confidence of final data in accordance with MTP 3-1-002. Provisions shall be made for modification during test progress as may be indicated by monitored test results.

j. Ensure that appropriate security measures are instituted to safeguard classified materiel and data, as applicable, and that arrangements for supporting and participating agencies, activities, and facilities have been made.

6.2 TEST CONDUCT

NOTE: Modification of these procedures shall be made as required by technical design of the item under test and availability of test equipment, but only to the extent that such modifications will not affect the validity of the test results.

6.2.1 Response Requirements

a. Consult references 4. F, G, and H, and prepare methods to evaluate the response requirements of the item under test.

b. Perform a pre-test reference calibration and checkout on the system and subsystems under test in accordance with the methods prepared above (Step a.).

c. Record the following:

1) Description of methods employed
2) Control data obtained to be utilized later as reference criteria.
3) Difficulties or malfunctions encountered.
4) Photographed oscilloscope traces of all critical test points.

6.2.2 EMC-RFI

Subject the item under test to electromagnetic interference tests in accordance with MIL-STD-461 and MIL-STD-462.

NOTE: References 4. F through 4.J, which have a direct application on this sub-test, shall be consulted for applicable information.

6.2.3 SFERICS Response

a. Emplace the test item and expose it to thunderstorm activity.

b. Activate the unit to a fully operational state.

c. Monitor data points within the test item during the period of thunderstorm activity.

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d. With external sensors, measure the following:

1) Waveform and amplitude of the RF associated with the lightning stroke.
2) Relative intensity of light associated with the electrical discharge.

e. Record the following:

1) Number of lightning strokes to which the item under test was exposed.
2) Number of events rejected.
3) Number of events processed.
4) Number of false alarms.
5) Photographed oscilloscope traces of the waveform and amplitude of the RF associated with each lightning stroke.
6) Test item battery condition.
7) Distance from each lightning occurrence.

6.2.4 Nuclear Weapons Effects

NOTE: A pre-test reference calibration and checkout test shall be performed on the item under test, as outlined in paragraph 6.2.1, prior to the start of each phase of this sub-test.

6.2.4.1 Gamma Radiation

a. Expose the test item to the linear accelerator (LINAC) gamma radiation generator.

b. Place the required gamma dose in the LINAC in accordance with the dose requirements specified in the QMR.

c. Locate dosimeters around the item during exposure.

d. Record the following:

1) Performance of unit prior to test.
2) Serial number of LINAC generator.
3) Instrumentation data as required.
4) Radiation dose and duration of pulse.

6.2.4.2 Neutron Radiation

a. Expose the test unit to the steady-state neutron generator which produces the dosage required by the QMR.

b. Start the dosage at an order of magnitude below that specified in the QMR.

c. Terminate the doseage when the neutron radiation level equals the QMR specifications or when the device fails.

d. Record the following:

1) Performance of unit prior to test.
2) Serial number of neutron generator.
3) Instrumentation data as required.

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6.2.4.3 Fast Burst Reactor Effects

a. Position the test item around the reactor. (Use the flux field reports to determine the positioning of the test item).
b. Place sulfur (S_{32}) dosimeters on or near the test item to provide a quantitative measure of the neutron flux.
c. Convert the measured environment to equivalent results of the plutonium (Pu 239) values.
d. Record the following:

1) Performance of item under test prior to test.
2) Serial number of reactor.
3) Radiation dose and duration of pulse.
4) Instrumentation data as required.

6.2.4.4 Thermal Radiation

a. Place the test item in a solar furnace test chamber.
b. Maintain a time control on the furnace to obtain a time history simulation of a nuclear thermal pulse.
c. Determine the total thermal flux with calorimeter measurements.
d. Record the following:

1) Performance of test item prior to test.
2) Total flux in calories per square centimeter.
3) Photographs of thermal effects.
4) Performance data as required.

6.2.4.5 Air Blast

a. Subject the test item to a planar shock front explosion of a magnitude specified in the QMR.
b. Place the item in the stand-by mode and emplace tactically in the shock front path of the explosion in a position providing an impulse equal to the product of overpressure and duration not to exceed the impulse specifications of the QMR.
c. Record the following:

1) Overpressure and time duration at the test point.
2) Photographs of the test item before and after the test.
3) Results of operational checks before and after the tests.

6.2.5 Microbiological Test

a. Set up the test item in the normal stand-by mode (non-powered).
b. Subject the item under test to the applicable microbial resistance tests as described in MTP 5-2-584, adding additional representative bacteria and streptomycetes as required.
c. At the end of the 28 day test period, subject the test item to a visual inspection and performance check and record results.
d. After inspection, change the battery and expose the test item

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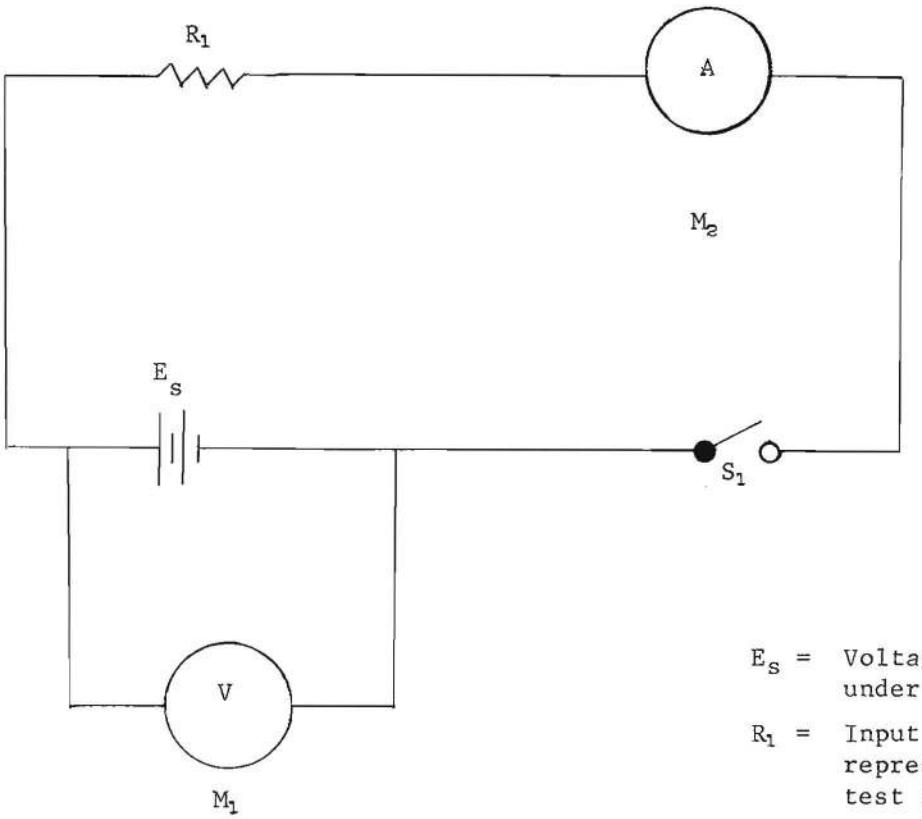
to the standard atmosphere for 24 hours.

e. Repeat Steps (b) and (c) above, and then return the test item to the normal stand-by mode.

6.2.6 Battery Load and Shelf Life Test

a. Connect the battery of the item under test to a switch, voltmeter, ammeter, and a resistive load whose input impedance is representative of the test item, as shown in Figure 1.

NOTE: This test circuitry shall be established and operated under normal laboratory environmental conditions.



E_S = Voltage of Battery under test.

R_1 = Input Impedance representative of test item.

M_1 = Voltmeter.

M_2 = Ammeter

S_1 = Switch

Figure 1. Battery Test Circuit

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b. Close the switch ("load" condition) and discharge the battery over an extended period of time as specified in the applicable QMR, or until failure of the battery.

c. Monitor the voltmeter and ammeter at least twice in each 24-hour period throughout the test.

d. Record the following:

1) Voltmeter readings.
2) Ammeter readings.
3) Elapsed time between meter readings.

e. Remove the test circuitry from the laboratory environment and repeat Steps (b), (c) and (d) above, under outside ambient conditions with the unit protected from direct natural elements but subjected to normal day - night temperature variations.

f. Repeat Steps (b), (c) and (d) above, with the test unit stored under extreme environmental conditions as specified in the applicable QMR.

g. Determine the life expectancy of the test item batteries as follows:

1) Store batteries under habitable room-ambient conditions.
2) Monitor twice weekly, for a maximum of six months, the "load" and "no-load" voltage until the batteries fail to meet the operating load criteria, and record results.

6.2.7 Equipment Reliability

a. Operate the equipment under test continuously for the period specified by the applicable QMR.

b. Monitor and record amount of servicing or maintenance required, if any, including battery changes.

c. Monitor and record number of false alarms, if any.

d. Determine and record the read-out rate (minutes) of the equipment after a nuclear (or simulated nuclear) event.

e. Repeat Steps (a) through (d) above, under various environmental conditions.

6.2.8 Air Drop

a. Instrument the item under test as required to measure impact shock in three mutually perpendicular planes.

b. Air drop the test item in accordance with the procedures outlined in MTP 7-2-509.

c. Record with the on-board instrumentation, from two locations in the test item, the resulting impact shock in the three mutually perpendicular planes.

d. After recovery, inspect the item for physical damage and subject the test item to operational tests as specified in paragraph 6.2.1, Response Requirements. Record results.

e. Repeat Steps (a) through (d) above, at least three times.

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6.3 TEST DATA

6.3.1 Preparation for Test

Data to be recorded prior to testing shall include but not be limited to:

a. Nomenclature, serial number(s), manufacturer's name, and function of the item(s) under test.
b. Nomenclature, serial number, accuracy tolerances, calibration requirements, and last date calibrated of the test equipment selected for the tests.
c. Damages to the test item incurred during transit and/or manufacturing defects.

6.3.2 Test Conduct

Data to be recorded in addition to specific instructions listed below for each sub-test shall include:

a. A block diagram of the test setup employed in each specified test. The block diagram shall identify by model and serial number, all test equipment and interconnections (cable lengths, connectors, attenuators, etc.) and indicate control and dial settings where necessary.
b. Photographs or motion pictures (black and white or color), sketches, charts, graphs, or other pictorial or graphic presentation which will support test results or conclusions.
c. An engineering logbook containing, in chronological order, pertinent remarks and observations which would aid in a subsequent analysis of the test data. This information may consist of temperatures, humidity, pressures, and other appropriate environmental data, or other description of equipment or components, and functions and deficiencies, as well as theoretical estimations, mathematical calculations, test conditions, intermittent or catastrophic failures, test parameters, etc., that were obtained during the test.
d. Test item sample size (number of measurement repetitions).
e. Instrumentation or measurement system mean error stated accuracy.

6.3.2.1 Response Requirements

Response requirements data to be recorded shall include:

a. Description of methods employed.
b. Control data to be utilized later as reference criteria.
c. Difficulties or malfunctions encountered.
d. Photographed oscilloscope traces of all critical test points.

6.3.2.2 EMC-RFI

EMC-RFI data shall be recorded as specified by MIL-STD-461 and 462.

6.3.2.3 SFERICS Response

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SFERICS response data to be recorded shall include:

a. Number of lightning strokes to which the test item was exposed.
b. Number of events rejected.
c. Number of events processed.
d. Number of false alarms.
e. Photographed oscilloscope traces of the waveform and amplitude of the RF associated with each lightning stroke.
f. Test item battery condition.
g. Distance from each lightning occurrence,

6.3.2.4 Nuclear Weapons Effects

6.3.2.4.1 Gamma Radiation

Record the following:

a. Performance of unit prior to test.
b. Serial number of LINAC generator.
c. Instrumentation data as required.
d. Radiation dose and duration of pulse.

6.3.2.4.2 Neutron Radiation

Record the following:

a. Performance of unit prior to test.
b. Serial number of neutron generator.
c. Instrumentation data as required.

6.3.2.4.3 Fast Burst Reactor Effects

Record the following:

a. Performance of test item prior to test.
b. Serial number of reactor.
c. Radiation dose and duration of pulse.
d. Instrumentation data as required.

6.3.2.4.4 Thermal Radiation

a. Record the following:

1) Performance of test item prior to test
2) Total flux in calories per square centimeter.
3) Performance data as required.

b. Retain photographs of thermal effects.

6.3.2.4.5 Air Blast

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a. Record the following:

1) Overpressure and time duration at the test point.
2) Results of operational checks before and after the test.

b. Retain photographs of the test item before and after the test.

6.3.2.5 Microbiological Test

Microbiological test data to be recorded shall consist of the results of the visual inspections and performance tests.

6.3.2.6 Battery Load and Shelf Life Test

Battery load and shelf life test data to be recorded shall include:

a. Voltmeter readings under load and no-load conditions.
b. Ammeter readings under load and no-load conditions.
c. Elapsed time between meter readings.

6.3.2.7 Equipment Reliability

Equipment reliability data to be recorded shall include:

a. Number and times of equipment failures, if any, for the period of continuous operation.
b. Amount of servicing or maintenance required, if any, including battery changes.
c. Number of false alarms, if any.
d. Equipment read-out rate (minutes) after a nuclear (or simulated nuclear) event.

6.3.2.8 Air Drop

Air drop data to be recorded shall include:

a. Impact shock for each drop.
b. Results of visual inspections and performance checks.

6.4 DATA REDUCTION AND PRESENTATION

Processing of raw test data shall, in general, consist of organizing, marking for correlation and identification, and grouping the test data according to sub-test title. Test criteria or test item specifications shall be noted on the test data presentation to facilitate analysis and comparison. Where necessary, test data measurements shall be converted to be compatible with units given by test criteria or specifications.

Specific instructions for the reduction and presentation of individual sub-test data are outlined in the succeeding paragraphs.

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6.4.1 Response Requirements

a. Battery voltage and current drain data shall be presented for any test in which the unit is activated.
b. Oscilloscope traces shall be labeled such that the traces can be easily interpreted, i.e., scope settings shall accompany each photograph.
c. Control data shall be presented in tabular form such that it provides ease of comparison to determine degradation of system performance.

6.4.2 EMC-RFI

a. Present the data as indicated by MIL-STD-461 and 462.
b. Compare the results of the test with the QMR requirements.
c. Compare the data and evaluate it for performance degradation.

6.4.3 SFERICS Response

a. Present the data in tabular form.
b. Compare the data acquired with the minimum requirements imposed by the QMR.
c. Report degradation in performance.

6.4.4 Nuclear Weapons Effects

a. Tabulate the data of each sub-test and identify properly with the corresponding test.
b. Present a tabulation of the performance data of the test item before and after the test.
c. Present a tabulation and summarize the effects of a nuclear detonation on the test item.
d. Photographs shall be presented, properly labeled, to show physical damage to the equipment.

6.4.5 Microbiological Test

a. Present in tabular form the results of the performance tests and the visual inspection.
b. Tabulate the results opposite the corresponding cycle.

6.4.6 Battery Load and Shelf Life Test

a. Tabulate all data and present in a manner suitable for comparison between data acquired and QMR requirements.
b. Determine from the data the reliability of the batteries and their life expectancy.

6.4.7 Equipment Reliability

Reduce and present the data in accordance with MTP 6-2-503.

6.4.8 Air Drop

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a. Present in tabular form the impact shock data and the results of the operational tests.

b. Tabulate all results opposite the corresponding air drop.

c. Compare the performance of the test item before and after the air drop to determine the system or subsystem degradation.

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